



Combined exposure of heat stress and ozone enhanced cognitive impairment via neuroinflammation and blood brain barrier disruption in male rats

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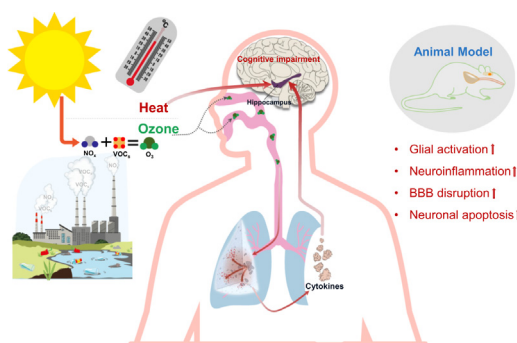
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HIGHLIGHTS

- HS induced cognitive decline and neuronal damage, and O₃ exposure exacerbated HS-induced neurocognitive toxicity.
- HS or/and O₃ contributed to neurocognitive impairment through glial activation-mediated neuroinflammation.
- BBB disruption might be an important convergence in aggravated neurotoxicity from multiple environmental exposures.

GRAPHICAL ABSTRACT



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ABSTRACT

Background: Heat stress (HS) exposure has been linked to cognitive dysfunction. In reality, high temperature does not occur alone in environment, and ozone (O₃) and heatwaves usually co-exist in atmospheric environment. However, whether O₃ exposure exacerbates HS-induced cognitive impairment and the potential underlying mechanisms have not been explored experimentally. The aim of this study was to determine the co-effects and mechanisms of HS and O₃ on the cognitive dysfunction.

Methods: 48 Sprague Dawley male rats were randomly divided into 4 groups: control, HS, O₃ and HS plus O₃ (HO₃) groups. Rats in HS and HO₃ group were exposed to 40 °C every morning from 9:00 to 12:00 for 15 consecutive days. While rats in O₃ and HO₃ groups were exposed to 0.7 ppm O₃ the same day from 14:00 to 17:00 for 15 days. Cognitive performance was examined with Morris water maze test. Neurodegeneration, glial activation, neuroinflammation, blood brain barrier (BBB) disruption and apoptosis were evaluated by Western blot, Elisa, immunohistochemistry and immunofluorescence staining.

Results: HS induced cognitive decline and neuronal damage in rats. Further studies showed that exposure of rats to HS could also induce glial activation, neuroinflammation and neuronal apoptosis in hippocampus, and decrease in the expressions of ZO-1, claudin-5 and occluding, indicative of BBB disruption. Impressively, the neuronal effects induced by HS, as depicted above, could be worsened by co-exposure to O₃ in rats.

Conclusions: Co-exposure to O₃ promotes HS-induced cognitive impairment in rats possibly through glial-mediated neuroinflammation and BBB disruption.

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1. Introduction

Heatwaves are one of the most dangerous natural hazards, but receive less attention because their effects are not always immediately apparent (World Health Organization, 2017). Heatwaves are increasing in their duration, frequency and magnitude due to the change of climate. Globally, population exposed to heat increased by 125 million or so between 2000 and 2016 (World Health Organization, 2018). Extreme ambient temperature may have critical impact on body's thermoregulation, physiological functions and behaviors (Nakamura et al., 2022). Many previous studies have demonstrated that exposure to heat stress (HS) causes multiple detrimental health effects including water loss (El Khayat et al., 2022), energy intake and energy expenditure (Moellering and Smith, 2012), sleep disturbance (Kim et al., 2020), mental health (Arifwidodo and Chandrasiri, 2020; Tawatsupa et al., 2010), heat stroke (Epstein and Yanovich, 2019), and cardiovascular diseases (Loughnan et al., 2010; Yang et al., 2018). In addition, the central nervous system (CNS) is extremely sensitive to HS (Kiyatkin, 2010). Several studies have shown that HS has a profound influence on brain structure and function, such as neuronal loss, brain atrophy, spasms, neurogenic deficits and cognitive impairment (Cedeño Laurent et al., 2018; Chauhan et al., 2021; Sinha, 2007). Given the fact of global warming, heat-related brain health effects have attracted much attentions from researchers.

Meanwhile, high ambient temperature promotes the production of greenhouse gas ozone (O_3) (National Research Council, 1991), a product of photochemical reaction among nitrogen oxides (NO_x) and volatile organic compounds (VOCs). High O_3 concentrations often occur following heat waves (Fiore et al., 2015; Lyu et al., 2022; Zhu et al., 2022). Concentrations of ground-level O_3 have been rising in the last few decades (Gaudel et al., 2020) because of global warming (Archer et al., 2019) and increasing emissions of NO_x and VOCs (Wang et al., 2019). Hence, the body exposed to high temperature does not occur alone in real world, rather co-exposes to other environmental factors such as O_3 . Recent epidemiological studies have demonstrated that O_3 exposure is associated with some human diseases, such as pulmonary impairment (Holm and Balmes, 2022; Niu et al., 2022), cardiovascular dysfunction (Chen et al., 2022; Raza et al., 2018) and even perturbation of brain function (Gao et al., 2022). Moreover, documented studies have indicated that the co-exposure to HS and O_3 may exacerbate health damage (Chen et al., 2018; Jhun et al., 2014). Although the effects of HS or O_3 alone on cognitive ability have been observed, studies on the co-effects of O_3 and HS on cognitive performance have been rarely explored. Ambient temperature and ground-level O_3 concentration are experiencing a continually stable increase, which render a new global health concern and warrant urgent investigation.

However, the underlying mechanisms underlying HS or O_3 co-exposure-induced human brain health are poorly understood. HS is an environmental stressor and closely related to the neuroimmune response (Haykin and Rolls, 2021; Lee et al., 2015; Quinteiro-Filho et al., 2012). Neuroinflammation has been used to explain cognitive impairment caused by HS (Lee et al., 2015; Sun et al., 2021). In addition, O_3 cannot directly reach the brain, suggesting the involvement of peripheral immune response to HS and O_3 in the pathogenesis of brain. Indeed, a recent study revealed that O_3 -induced peripheral immune response may be indirectly implicated in Alzheimer's disease through the lung-brain axis (Greve et al., 2022). However, the findings from this study need further verification. Toxicological evidence concerning co-exposure to HS and O_3 is scarce and the underlying mechanism remains unknown. Therefore, it is imperative to systematically explore the toxicological effects of HS and O_3 exposure.

In this study, neurobehavioral, molecular biology, and pathological approaches were administered to mainly investigate the toxicological effects of co-exposure to HS or O_3 on cognitive function. Moreover, the underlying toxicological mechanism of exposure to HS and O_3 on cognitive function was explored.

2. Materials and methods

2.1. Animal treatment

Male Sprague Dawley (SD) rats (200–220 g) were housed in steel cages under the controlled environmental condition (temperature 22–23 °C, relative humidity 35–55 % and 12-h light/dark cycle). Food and drinking water was provided to rats ad libitum. After 1 week of acclimation, 48 SD rats were randomly divided into 4 groups: control, heat stress (HS), ozone (O_3) and HS plus O_3 (HO_3) groups ($n = 12$). HS and HO_3 rats were exposed to 40 °C heat every morning from 9:00 to 12:00 for 15 consecutive days. While O_3 and HO_3 rats were exposed to 0.7 ppm O_3 the same day in the afternoon from 14:00 to 17:00 for 15 consecutive days. A previous study has shown that 2 weeks of heat stress exposure can induce memory impairment in mice (Lee et al., 2015). And the duration of extreme high temperature (>35 °C) usually lasted 2 weeks or longer in realistic environment (Fig. S1). Therefore, the duration of exposure was set up for 15 days. The temperature (40 °C) and O_3 concentration (0.7 ppm) in exposure chambers were set up referred to previous studies (Lee et al., 2015; Nery-Flores et al., 2018). The interval between heat and subsequent exposure to O_3 was 2 h. Body weight was measured at the onset of exposure and monitored every 7 days. Body core temperature was measured at 12:00 on the last day of exposure. To obtain rectal temperature, rats were hand-restrained and placed on a flat platform. The tail was lifted and a rectal temperature probe (covered with Vaseline) was inserted into the rectum to a fixed depth (up to 4 cm). All animal procedures were approved by Ethics Committee of Hainan Medical University, and performed in accordance with National Institutes of Health (NIH) Guide for the Care and Use of Laboratory Animals.

2.2. Morris water maze

Morris water maze (MWM) was performed to evaluate the capacity of cognition in rats. The MWM test was performed in a circular swimming pool (diameter 90 cm and height 40 cm), which was filled with water and maintained at a temperature of 25 °C. As shown in Fig. 2A, a hidden platform was located in the third quadrant below the water surface (1 cm). The position of the platform is fixed throughout the training and testing process. There are four visual black cues of different shapes around the swimming pool. This experiment contained two-phase training protocol: navigation and probe tests. After the first day of training, a navigation test was conducted on the second to fifth days. Rats were randomly placed into one of the quadrants and allowed freely to find the platform up to 60 s. We recorded the time and traveled distance to find the platform. All rats were trained 4 times per day (corresponding to four quadrants). The average value of the four times training data was used to assess the learning performance of rats on that day. After taking a day off, the probe test was performed on the seventh day. The platform was removed from the pool. All rats were placed into the swimming pool from the second quadrant and allowed to freely navigate in the pool for up to 60 s. We recorded the time rats crossed the platform for the first time and the number of times it crossed the platform within 60 s. Different indexes of cognition (i.e., traveled distance, escape latency, swimming speed, latency for first platform crossing and number of platform crossing) and swimming paths were recorded by a video system and analyzed using EthoVisionXT13.

2.3. Brain tissue preparation and treatment

Rats were anesthetized by intravenous administration of isoflurane and perfused with 4 % paraformaldehyde (PFA) followed by whole brain harvesting and processing. After 24 h of post-fixation in 4 % PFA, the brain tissues were embedded in paraffin and sectioned (4 μ m).

2.4. Histopathology examination of hippocampus

Histopathology of hippocampus was assessed by Hematoxylin/eosin (H&E) staining and Nissl staining. Paraffin sections were deparaffinized

according to standard protocols. For H&E staining, the sections were stained with hematoxylin solution and eosin solution for 5 min, respectively. Then the sections were dehydrated, mounted and observed under the light microscope (TE2000, Nikon, Japan). For Nissl staining, the dewaxed sections were stained with toluidine blue solution for 5 min and differentiated in 1 % glacial acetic acid. Subsequently, sections were dehydrated and mounted for observing.

2.5. Immunofluorescence staining

The sections were deparaffinized and probed with antibodies against claudin-5 (Invitrogen, USA), ionized calcium binding adaptor molecule-1 (Iba-1, Wako Chemicals, USA), glial fibrillary acidic protein (GFAP) and caspase-3 (Abcam, USA) overnight at 4 °C. On the second day, the sections were washed and then incubated with biotin-labeled or Alexa-488/594 conjugated secondary antibodies at room temperature for 60 min in the dark. Olympus microscope was used to acquire images. ImageJ was used

to quantify the staining density of claudin-5, Iba-1, GFAP and caspase-3 in the hippocampus.

2.6. Western blot analysis

Western blot was performed to explore the effects of different exposures on protein expressions. Lysis of hippocampal tissue was performed in RIPA buffer following the manufacturer's instructions. BCA kit was used to detect the protein concentrations. Subsequently, equal amount of protein in four groups were analyzed by sodium dodecyl sulfate polyacrylamide gel electrophoresis (SDS-PAGE), followed by transfer to the polyvinylidene fluoride (PVDF) membrane. After blocking with 5 % non-fat milk, the membrane was probed with primary antibodies against postsynaptic density protein 95 (PSD95, Abcam, USA), Neuronal nuclear protein (Neun, Abcam, USA), Iba-1 (Abcam, USA), GFAP (Abcam, USA), interleukin-1 β (IL-1 β , Abcam, USA), Tumour necrosis factor α (TNF- α , Abcam, USA), caspase-3 (Abcam, USA), B-cell lymphoma protein 2 (BCL-2, Abcam, USA) and Bcl-2-associated X (BAX, Abcam, USA), matrix metalloproteinases-2

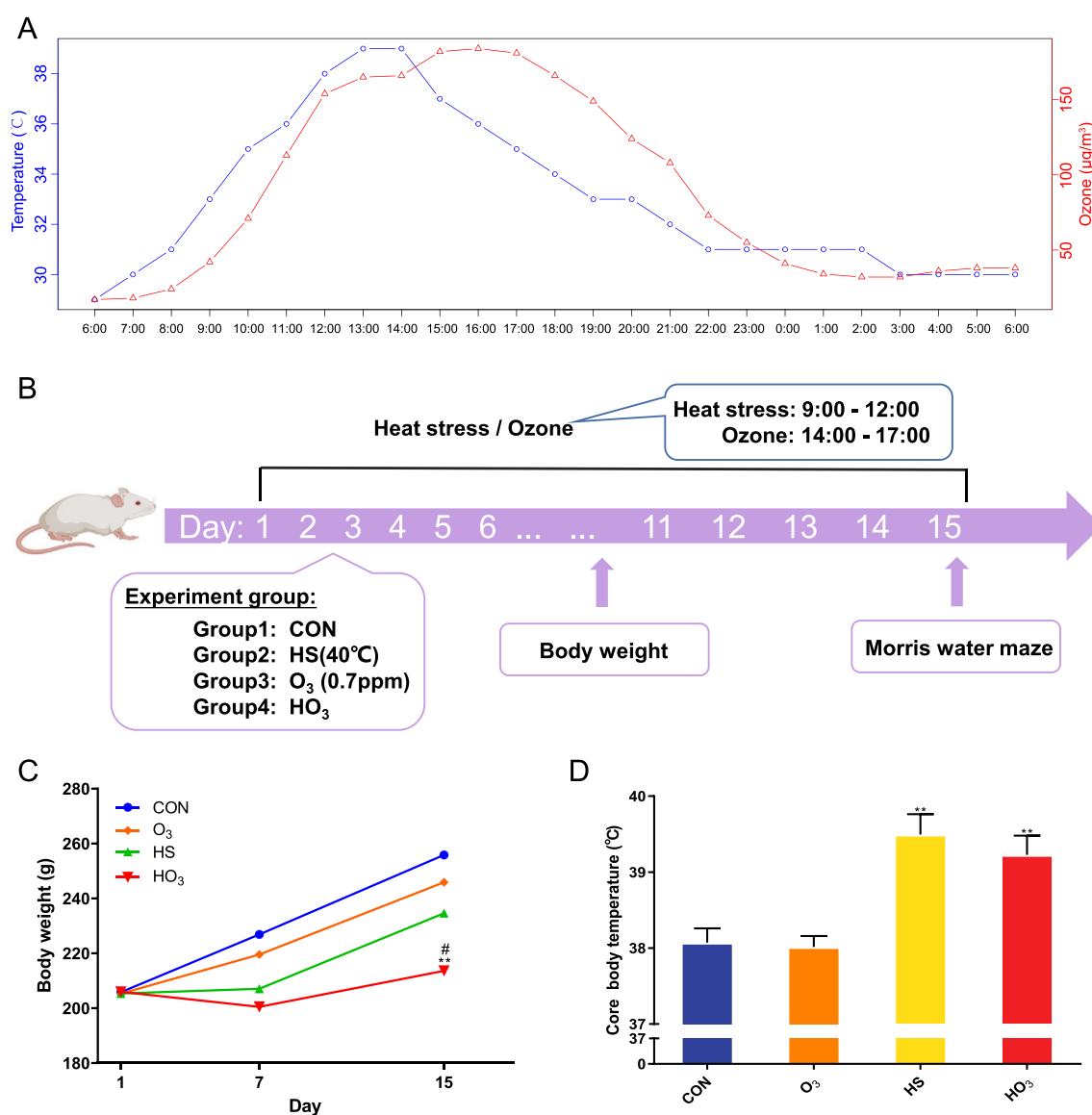


Fig. 1. Study design and effects of HS and O₃ exposure on body weight and core body temperature. (A) Curve of temperature and O₃ change on a certain day in a certain weather in Shanghai. (B) Schematic diagram of study design. (C) Change of body weight at the onset of exposure and monitored every 7 days. (D) Body core temperature on the last day of exposure. Data are presented as the mean $\pm$ SEM. * P < 0.05 and ** P < 0.01 versus CON group. # P < 0.05 and ## P < 0.01 versus HS group.

(MMP-2, Invitrogen, USA), matrix metalloproteinases-9 (MMP-9, Invitrogen, USA), claudin-5 (Invitrogen, USA), zonula occludens-1 (ZO-1, Invitrogen, USA), occluding (Invitrogen, USA) overnight at 4 °C. On next morning, the membrane was incubated with secondary antibody for 2 h at room temperature. The signals of blots were detected using ECL detection system and quantified with ImageJ software. GAPDH was considered as the internal control.

2.7. Statistical analysis

All results were expressed as mean $\pm$ standard deviation (SD). The differences between multiple groups were tested using one-way ANOVA, followed by Tukey post hoc test for pairwise comparisons at a 95 % confidence level. Meanwhile, two-way ANOVA was employed to determine HS and O₃ interactions. All analyses were performed using SPSS and R 4.1.1.

3. Results

3.1. Effects of HS and O₃ exposure on body weight and temperature

As shown in Fig. 1C, rats exposed to HS or/and O₃ gained weight slower than controls. After exposure for 15 days, the body weight of rats in HO₃ group was significantly lower than that of the control group (Fig. 1C). Moreover, core body temperature was remarkably increased in rats in HS and HO₃ group compared to controls (Fig. 1D).

3.2. O₃ exposure exacerbated HS-induced cognitive decline

MWM test was performed to determine whether exposure to HS and O₃ could impair cognitive capacity. Spatial learning ability was shown in Fig. 2B and C. No significant difference was observed between

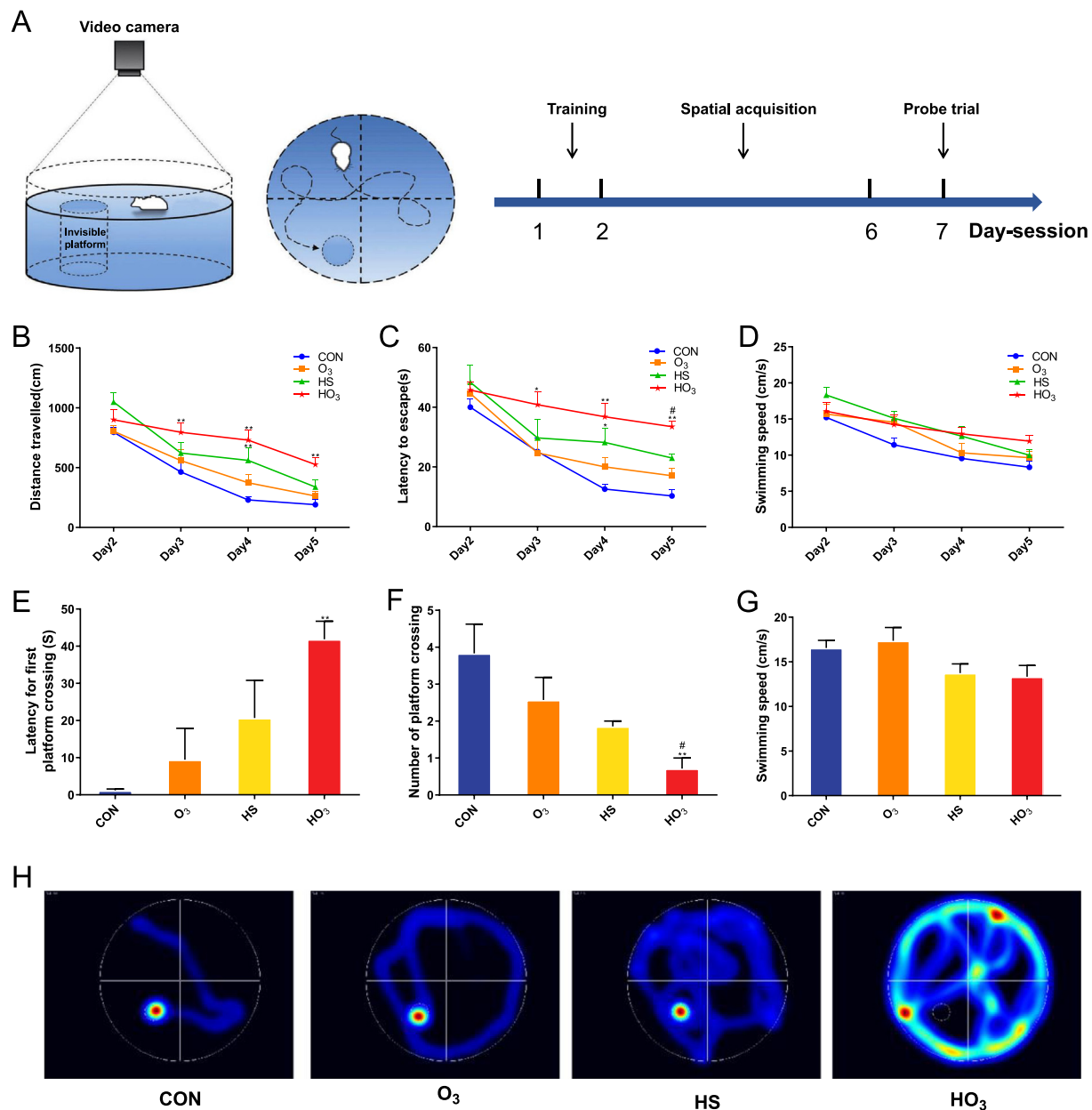


Fig. 2. Changes in spatial learning and memory ability of rats at 15 days after exposure to HS and O₃. (A) Schematic diagram of the operation of Morris water maze test. (B) Changes in traveled distance of navigation test. (C) Changes in escape latency of navigation test. (D) Changes in swimming speed of navigation test. (E) Latency for first platform crossing in probe trial. (F) The number of crossing platform in probe trial. (G) Swimming speed in probe trial. (H) Representative images of typical moving trajectory in probe trial. Data are presented as the mean $\pm$ SEM. * P < 0.05 and ** P < 0.01 versus CON group. # P < 0.05 and ## P < 0.01 versus HS group. The first platform crossing and reduced the number of crossing platform.

O₃-exposed rats and controls in escape latency and traveled distance. HS exposure significantly elevated escape latency and traveled distance on the fourth day. However, rats exposed to HO₃ showed a longer time to locate the platform and increased traveled distance compared to

controls from the third day onwards. The probe test revealed that rats in HO₃ group displayed elevated latency for the first platform crossing and reduced number of crossing platform compared with control group (Fig. 2E–F), indicating impaired memory ability. In addition,

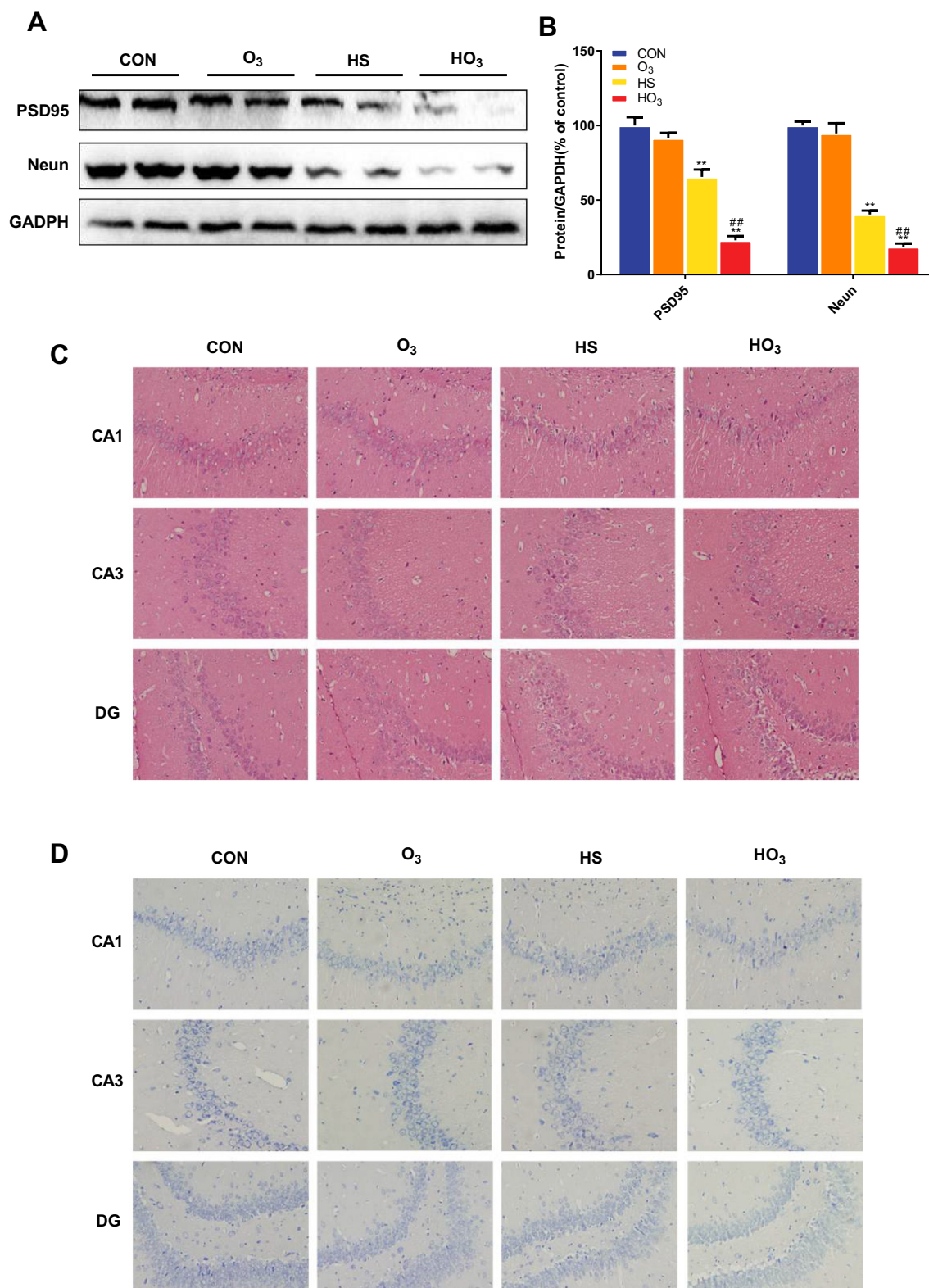


Fig. 3. Effects of HS and O₃ exposure on hippocampal neuron damage and microstructure of rats. (A) Representative western blot bands of PSD95 and Neun. (B) Protein quantification of normalized PSD95 and Neun. (C) Representative histopathology of the hippocampal CA1, CA3 and DG regions in rats by H&E staining at 400 × magnification. Data are presented as the mean ± SEM. *P < 0.05 and **P < 0.01 versus CON group. #P < 0.05 and ##P < 0.01 versus HS group.

rats in HO₃ group showed decreased number of crossing platform compared with rats in HS group (Fig. 2F). Fig. 2H shows the representative traveled trajectories of rats finding the platform on the fifth day. Noticeably, rats in control group mainly swam in the central areas of the pool, and searching the platform around the areas where it previously existed. In contrast, rats in exposure group mainly swam around the pool without searching for the platform. There was no significant difference in swimming speed for all rats (Fig. 2D and G), suggesting that the differences in MWM test among groups were not due to locomotor deficits.

3.3. O₃ exposure aggravated hippocampal neuron damage in HS-exposed rats

Next, the disruption of synaptic protein expression was examined in response to HS and O₃ exposure (Fig. 3A–B). Compared with the control group, HS exposure decreased the expression levels of PSD95 and Neun in the hippocampus by 34.42 % and 59.68 %, respectively. Additionally, the protein levels of PSD95 and Neun were significantly reduced by 77.02 % and 81.39 %, respectively, in rats in HO₃ group compared with rats in HS group. Post hoc analyses showed PSD95 and Neun in the hippocampus were significantly decreased in HO₃ group compared with HS group. Further interaction plots showed a significant co-effect between HS and O₃ with respect to aforementioned indicators (Fig. S3).

H&E and Nissl staining showed that neurons in hippocampal regions were regularly and tightly arranged with obvious nucleus and no noticeable cell loss in control group, while neurons in CA1, CA2 and DG hippocampal regions in HS group were loosely arranged with blurred cell border, increased nuclear pyknosis and inconspicuous nuclei (Fig. 3C and D). Intriguingly, the combined exposure to O₃ and HS exacerbated these deformations. Nissl bodies stained blue in Nissl staining dissolved and even disappeared in pathological conditions. Compared with control group, the neurons in the HS group were light staining, indicative of the presence of fewer Nissl body. Further, more lighter staining cells were observed in the combined exposure group (Fig. 3D).

3.4. O₃ exposure exacerbated HS-induced glial activation and neuroinflammation

Iba-1 and GFAP were two biomarkers for microglia and astrocyte activation, respectively. Immunofluorescence staining was used to investigate whether glial activation contributed to HO₃-induced cognitive deficits in rats. Fluorescence staining with Iba-1 antibody showed that microglia in the hippocampus of the control rats displayed ramified morphology, while microglia in exposure rats presented enlarged cell body and hypertrophied morphology, indicators of activated microglia (Fig. 4A). GFAP immunostaining showed that astroglial cells in exposure groups displayed hypertrophy morphology and intensified GFAP staining density, indicative of astroglial activation (Fig. 4A). Quantitative analysis revealed increased numbers of Iba-1⁺ and GFAP⁺ cells in the hippocampus of HS-exposed rats compared with controls (Fig. 4B–C). Furthermore, these changes were exacerbated in rats co-exposed to HS and O₃ (Fig. 4B–C). Activation of microglia and astrocytes in the hippocampus was further confirmed by Western blot, showing higher expression levels of Iba-1 and GFAP in three exposure groups than controls (Fig. 4D–E). The protein expression levels of Iba-1 in rats in O₃, HS and HO₃ groups were 1.81-, 2.35- and 3.15-fold higher than that in control group, respectively (Fig. 4D–E). Compared to control rats, protein levels of GFAP were 1.55-, 1.98- and 2.56-fold higher in rats exposed to O₃, HS and HO₃, respectively (Fig. 4D–E). In addition, co-exposure to HS and O₃ enhanced this effect in the hippocampus compared to exposure to either HS or O₃.

To further determine whether glial activation was required for the up-regulation of proinflammatory factors, the proteins expression levels of IL-1 β and TNF- α in the hippocampus of rats were measured. Levels of IL-1 β in O₃, HS and HO₃ groups were significantly increased by 1.72-fold, 2.25-fold and 3.92-fold compared to that in control group, respectively (Fig. 4F–G), while the levels of TNF- α in rats exposed to HS and HO₃

were 3.58-fold and 4.90-fold higher than that in control group, respectively (Fig. 4F–G). Post-hoc analyses showed significant difference in aforementioned indicators between HO₃ group and HS group. The results from this study also revealed a significant co-effect between HS and O₃ exposure on IL-1 β expression in the hippocampus (Fig. S4E).

3.5. Co-exposure aggravated neuronal apoptosis in the hippocampus

It is well documented that Bcl-2 and Bax are closely related to anti-apoptotic and proapoptotic functions, respectively, and caspase-3 is the main executor of apoptosis (Zhong et al., 2018). As shown in Fig. 5A–B, exposure to HS and HO₃ increased apoptosis in hippocampal neurons, represented as 3.22-fold and 5.39-fold increase in caspase-3 expression than that in control group, respectively. In addition, the Bcl-2/Bax ratios of rats in HS and HO₃ groups decreased to 60.56 % and 24.38 % of that in control group, respectively (Fig. 5A and C). This effect was more obvious in HO₃ group than in HS group. Moreover, the apoptosis of hippocampal neurons was confirmed by Immunofluorescence, with higher caspase-3 expression in HO₃ group than in controls (Fig. 5D). Furthermore, the interaction plots demonstrated a significant co-effect between HS and O₃ in caspase-3 expression and Bcl-2/Bax ratios (Fig. S5).

3.6. O₃ exposure enhanced peripheral inflammation and elevated permeability of BBB in HS-exposed rats

ELISA results showed that O₃ exposure significantly increased serum IL-1 β and TNF- α levels by 1.5-fold and 2.1-fold, respectively, compared to the control, indicating the occurrence of peripheral inflammation (Fig. 6A). MMP2 and MMP9 are two biomarkers for degradation of extracellular matrix and tight junction (TJ) proteins. As illustrated in Fig. 6C–D, the expression of MMP2 in the hippocampus of rats in HS group was 2.02-fold high than that in control group. Compared with control, HO₃ exposure increased the expression levels of MMP2 and MMP9 by 3.17-fold and 3.05-fold, respectively (Fig. 6C–D). Similarly, co-exposure to HS and O₃ induced more expression of MMP2 and MMP9 than HS exposure alone (Fig. 6C–D). Three TJ proteins, including ZO-1, claudin-5 and occluding, were key players to maintain the functional property of BBB (Sweeney et al., 2018). Western blot showed that exposure to HS and HO₃ reduced the expressions of TJ proteins in the hippocampus compared with control (Fig. 6E–F). HS exposure decreased expression levels of ZO-1, claudin-5 and occluding in the hippocampus by 33.72 %, 38.40 % and 51.05 %, respectively, relative to control (Fig. 6E–F). Impressively, the protein levels of ZO-1, claudin-5 and occluding were significantly decreased by 62.61 %, 63.85 % and 81.08 %, respectively, by exposure to HO₃ compared with HS exposure (Fig. 6E–F). We also observed that combined exposure to HS and O₃ enhanced this effect compared to a single exposure to HS (Fig. 6E–F). Further analysis revealed a significant co-effect between HS and O₃ on the expression of MMP-2, ZO-1 and claudin-5 proteins (Fig. S6). Immunostaining assay using antibodies against claudin-5 also confirmed that HO₃ exposure decreased the expressions of TJ proteins (Fig. 6B).

4. Discussion

To our knowledge, this present study first revealed that O₃ exacerbated HS-induced cognitive impairment by neuroinflammation and worsening BBB damage. The main findings from this study include: (1) HS induced cognitive decline and neuronal damage, and O₃ exposure exacerbated neurocognitive impairment induced by HS; (2) O₃ enhanced HS-induced microglia and astrocyte activation and subsequent neuroinflammation; (3) O₃-induced peripheral inflammation promotes HS-induced increase in the permeability of BBB.

MWM test, one of the most commonly used models to measure spatial learning and memory in rodents (Lee et al., 2022; Vorhees and Williams, 2006; Yu et al., 2020), were performed to assess the effect of HS or/and

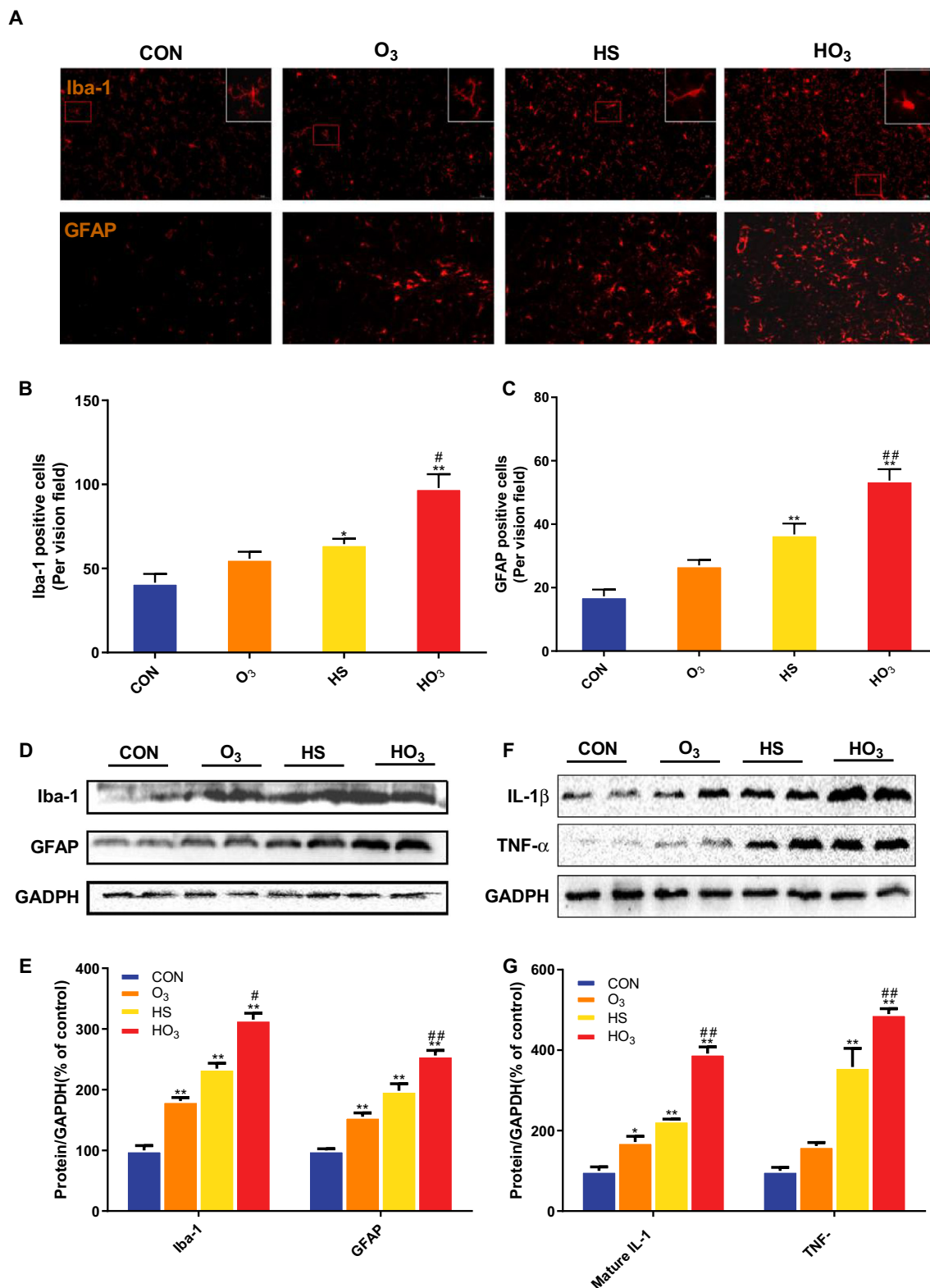


Fig. 4. Effects of HS and O₃ exposure on hippocampal glia activation and neuroinflammation of rats. (A) Representative microphotograph of immunofluorescence staining showed the expression of Iba-1 and GFAP in the hippocampus. (B) Fluorescence quantification of normalized Iba-1. (C) Fluorescence quantification of GFAP. (D) Representative western blot bands of Iba-1 and GFAP. (E) Protein quantification of normalized Iba-1 and GFAP. (F) Representative western blot bands of IL-1 β and TNF- α . (G) Protein quantification of normalized IL-1 β and TNF- α . Data are presented as the mean $\pm$ SEM. * P < 0.05 and ** P < 0.01 versus CON group. # P < 0.05 and ## P < 0.01 versus HS group.

O₃ exposure on cognitive ability of rats. In line with the findings from previous studies (Chauhan et al., 2021; Lee et al., 2015), our results showed a worsening trend of spatial learning and memory in rats exposed to HS, and

this impairment was exacerbated by the following exposure to O₃. It is well recognized that the number and morphological features of hippocampal neurons are vital for learning and memory (Deng et al., 2010; Hanganu

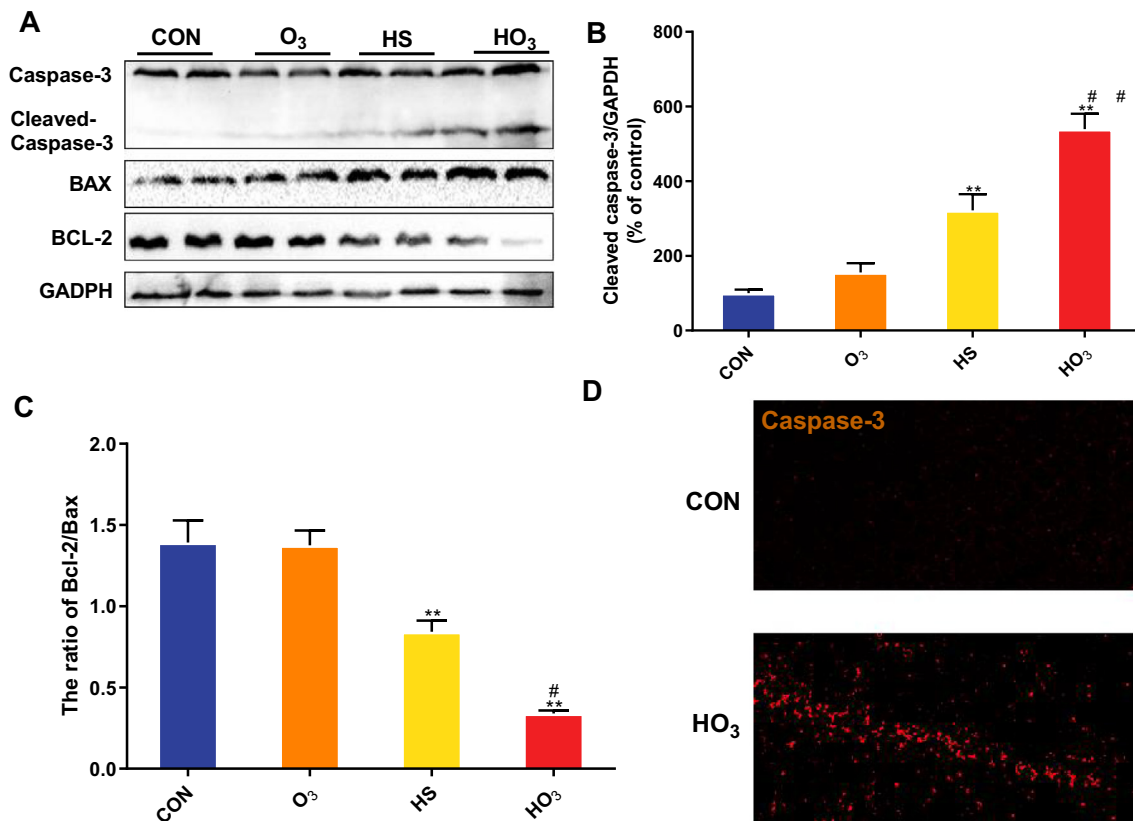


Fig. 5. Effects of HS and O₃ exposure on hippocampal neuronal apoptosis of rats. (A) Representative western blot bands of Caspase-3, Cleaved Caspase-3, BAX and BCL-2. (B) Protein quantification of normalized Cleaved Caspase-3. (B) Quantification of the ratio of BCL-2 and BAX. (D) Representative microphotograph of immunofluorescence staining showed the expression of Cleaved Caspase-3 in the hippocampus. Data are presented as the mean $\pm$ SEM. * P < 0.05 and ** P < 0.01 versus CON group. # P < 0.05 and ## P < 0.01 versus HS group.

et al., 2014). Neun is localized in nuclei of most of the neurons and commonly used as a specific biomarker for neurons (Wolf et al., 1996). PSD95 is a key postsynaptic protein (Kim and Sheng, 2004) involving in maintenance of the synaptic stability and plasticity of neurons (El-Husseini et al., 2000). Excitatory neurons change synaptic function through regulating expression of PSD95 (Diering and Haganir, 2018; El-Husseini et al., 2000; Liu et al., 2021). Reduced expression PSD95 is significantly associated with cognitive impairment in neurodegenerative disorders (Hou et al., 2019). Our findings from this present study demonstrated that HS exposure significantly decreased the number of hippocampal neurons and inhibited the expression levels of PSD95 protein in hippocampus, implying neuronal loss and damage. Importantly, these neurodegenerative alterations induced by HS were further worsened by following exposure to O₃. Thus, we assumed that neurocognitive impairment induced by HS could in part be attributed to neuronal and synaptic loss and this process can be exaggerated by subsequent exposure to O₃.

Currently, the mechanisms underlying the detrimental effect of HS and O₃ on cognition are still under speculation. Chronic neuroinflammation mediated by glial cells is a key contributor to the process of neurodegenerative (Che et al., 2018; Ronaldson and Davis, 2020). Glial cells, the innate immune cells of the brain parenchyma, have a critical role in regulating neuroinflammatory responses in the CNS and are activated in some neuropathological conditions (Glass et al., 2010). The activation of glia cells can lead to neuronal and synaptic damage through releasing pro-inflammatory cytokines, such as TNF- α and IL-1 β (Block et al., 2007; Glass et al., 2010; Zhang et al., 2022). Neuroinflammation has been negatively associated with cognitive decline (Che et al., 2018; Hou et al., 2019). A recent study reported that neuroinflammation led to neuronal loss and cognitive impairment by inducing apoptosis (Zhang et al., 2021). This was supported by the results from this study showing that HS induced

the activation of microglia and astrocytes in the hippocampus and associated increase in TNF- α and IL-1 β expression. Furthermore, this study also demonstrated that HS exposure caused apoptosis of hippocampal neurons, represented by elevated expression of cleaved caspase-3 and Bax as well as decreased expression of Bcl-2. It is noteworthy that inhalation of high concentrations of O₃ following HS exposure on the same day posed significant effects on glial activation, neuroinflammation, and neuronal apoptosis. As a whole, these results indicated that HS and O₃ may synergistically contribute to cognitive impairment and neuronal apoptosis through glial activation-mediated neuroinflammation.

The BBB is a highly selective semipermeable interface formed by endothelial cells that tightly controls the movement of cells, molecules and ions between the blood and the brain (Abbott et al., 2010; Blanchette and Daneman, 2015). The integrity of BBB is critical for maintaining normal physiology of the brain microenvironment and disruption of BBB plays an important role in the early stages of cognitive impairments in human (Nation et al., 2019). The results from this present study indicated that HS exposure reduced the expression of TJ proteins in hippocampus of rats, which could be implying increased BBB permeability. Moreover, this study also showed that BBB disruption induced by HS was accompanied by glial activation and neuroinflammation in rats. Not surprisingly, HS effect on BBB permeability was further strengthened by the following O₃ exposure. These observations were consistent with the notion that the activation of perivascular microglia and neuroinflammation were key events resulting in disruption of the BBB in stress and related illnesses (Haruwaka et al., 2019; Welcome and Mastorakis, 2020). Inflammation mediated by peripheral immune activation has been shown to disrupt the BBB in previous study (Huang et al., 2021). Specifically, a study demonstrated that the increasing levels of two inflammatory factors, such as IL-1 β and TNF- α , in the periphery are primarily associated with activation of

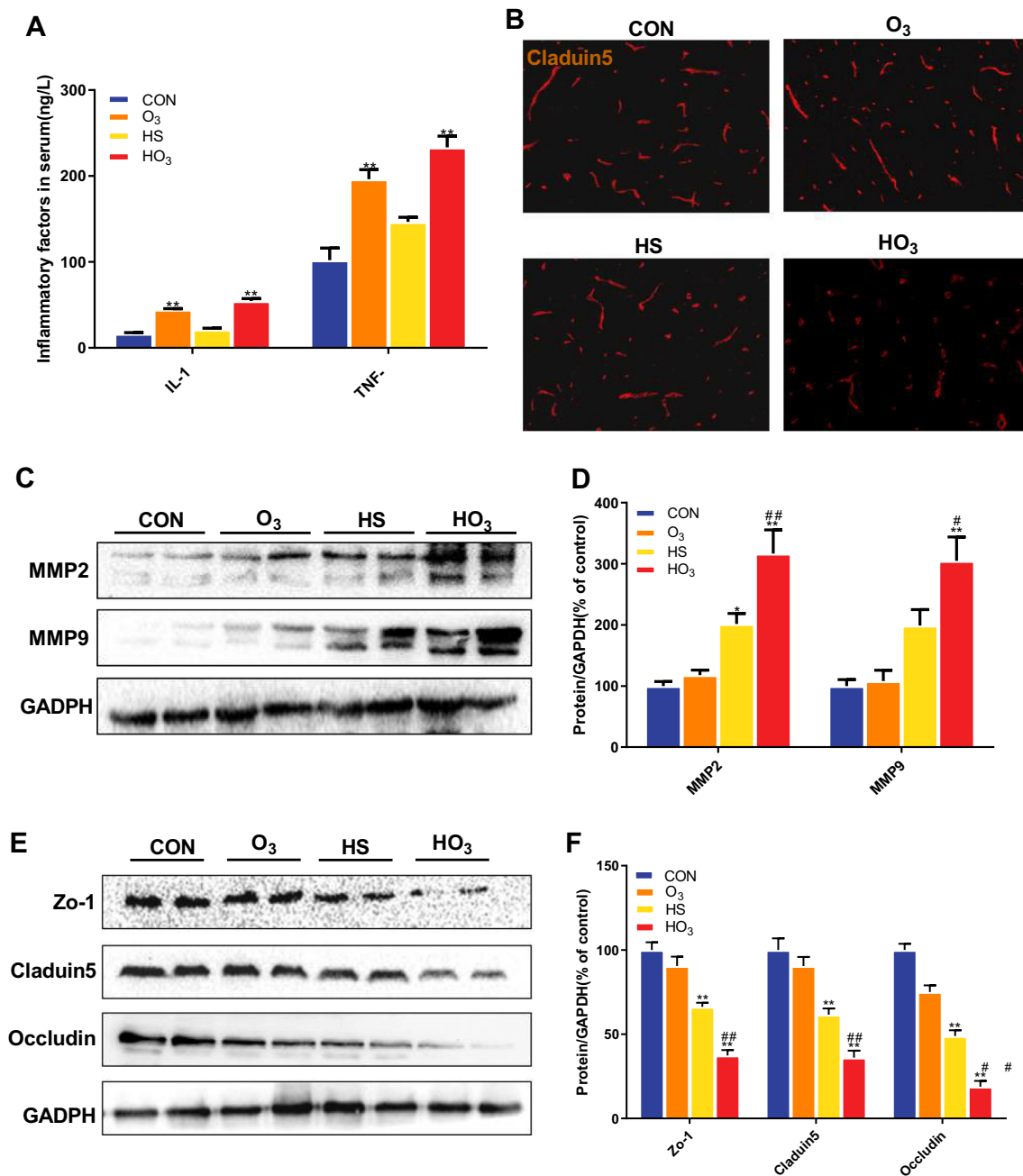


Fig. 6. Effects of HS and O₃ exposure on peripheral inflammation and blood brain barrier of rats. (A) The content of IL-1 β and TNF- α in serum. (B) Representative microphotograph of immunofluorescence staining showed the expression of Claduin-5 in the hippocampus. (C) Representative western blot bands of MMP-2 and MMP-9. (D) Protein quantification of normalized MMP-2 and MMP-9. (E) Representative western blot bands of ZO-1, Claduin-5 and Occludin. (F) Protein quantification of normalized ZO-1, Claduin-5 and Occludin. Data are presented as the mean $\pm$ SEM. * P < 0.05 and ** P < 0.01 versus CON group. # P < 0.05 and ## P < 0.01 versus HS group.

peripheral innate immunity (Dinarello, 2018). Other studies also demonstrated that BBB disruption contributes to environmental exposure-induced neurotoxicity (Disdier et al., 2017) or neurocognitive disorder by releasing vasculature-derived substances (Sweeney et al., 2018). In support of these observations, the results from this study showed that O₃ exposure increased the levels of pro-inflammatory factors including IL-1 β and TNF- α in circulation, BBB disruption, neuroinflammation, ultimately adverse effects on brain function.

Thus far, the mechanisms by which co-exposure to HS and O₃ affect cognitive function have not been elucidated. Based on the findings from this study, we assume that HS-induced production of pro-inflammatory factors from peripheral response and glia activation may stimulate BBB

disintegration, resulting in neuronal apoptosis and accelerated cognitive impairment. Inflammatory cytokines in circulating induced by O₃ further damage the BBB, leading to damage factors and enables toxic blood-derived molecules to enter the brain, resulting in more severe neurotoxicity and neurocognitive damage (Fig. 7).

5. Conclusions

In summary, the results from this study indicate that O₃ exposure promotes HS-induced cognitive impairment through peripheral inflammation, enhanced glial activation-mediated neuroinflammation, BBB damage, and neuronal apoptosis. Therefore, this study emphasizes the necessity to

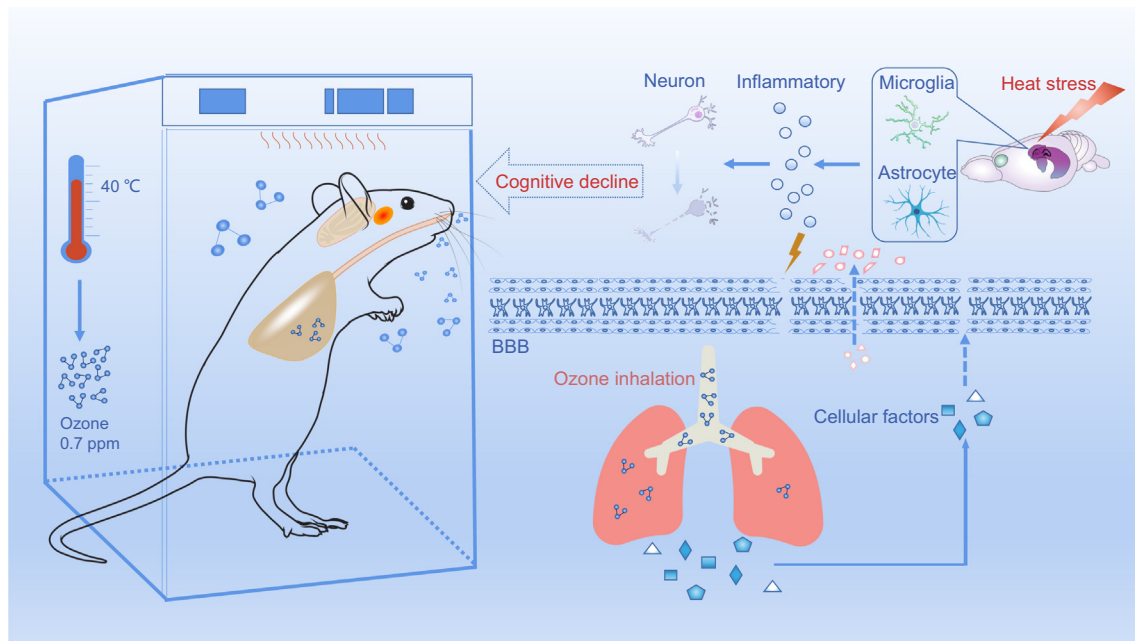


Fig. 7. Schematic diagram showed the possible mechanisms of cognitive decline induced by HS and O₃ exposure. O₃ may aggravate HS-induced cognitive impairment by enhancing glial activation-mediated neuroinflammation, BBB damage, and neuronal apoptosis. The BBB damage was served as an important node and implicated in cognitive impairment induced by co-exposure of HS and O₃.

evaluate the effects of mixed exposures on brain health in realistic environment. Furthermore, our findings highlight that BBB disruption may be an important convergence in aggravated neurotoxicity from multiple environmental exposures. However, the potential underlying mechanisms remain unclear and available neuropathology evidence warrant further experimental and epidemiological verification.

CRedit authorship contribution statement

ZY and LBZ: Conceptualization; WDW: Investigation; YML and YHJ: Formal analysis and Visualization; ZY: Writing—original draft; LBZ: Writing—review & editing. All authors have read and agreed to the published version of the manuscript.

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Data availability

All original data are available from the corresponding author upon request.

Declaration of competing interest

None.

Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.scitotenv.2022.159599>.

References

Abbott, N.J., Patabendige, A.A., Dolman, D.E., Yusof, S.R., Begley, D.J., 2010. Structure and function of the blood-brain barrier. *Neurobiol. Dis.* 37, 13–25.

- Archer, C.L., Brodie, J.F., Rauscher, S.A., 2019. Global warming will aggravate ozone pollution in the U.S. mid-Atlantic. *J. Appl. Meteorol. Climatol.* 58, 1267–1278.
- Arifwiddo, S.D., Chandrasiri, O., 2020. Urban heat stress and human health in Bangkok, Thailand. *Environ. Res.* 185, 109398.
- Blanchette, M., Daneman, R., 2015. Formation and maintenance of the bbb. *Mech. Dev.* 138 (Pt 1), 8–16.
- Block, M.L., Zecca, L., Hong, J.S., 2007. Microglia-mediated neurotoxicity: uncovering the molecular mechanisms. *Nat. Rev. Neurosci.* 8, 57–69.
- Cedeño Laurent, J.G., Williams, A., Oulhote, Y., Zanobetti, A., Allen, J.G., Spengler, J.D., 2018. Reduced cognitive function during a heat wave among residents of non-air-conditioned buildings: an observational study of young adults in the summer of 2016. *PLoS Med.* 15, e1002605.
- Chauhan, N.R., Kumar, R., Gupta, A., Meena, R.C., Nanda, S., Mishra, K.P., et al., 2021. Heat stress induced oxidative damage and perturbation in bdnf/erk1/2/creb axis in hippocampus impairs spatial memory. *Behav. Brain Res.* 396, 112895.
- Che, Y., Hou, L., Sun, F., Zhang, C., Liu, X., Piao, F., et al., 2018. Taurine protects dopaminergic neurons in a mouse Parkinson's disease model through inhibition of microglial m1 polarization. *Cell Death Dis.* 9, 435.
- Chen, K., Wolf, K., Bretnier, S., Gasparrini, A., Stafoggia, M., Samoli, E., et al., 2018. Two-way effect modifications of air pollution and air temperature on total natural and cardiovascular mortality in eight European urban areas. *Environ. Int.* 116, 186–196.
- Chen, R., Jiang, Y., Hu, J., Chen, H., Li, H., Meng, X., et al., 2022. Hourly air pollutants and acute coronary syndrome onset in 1.29 million patients. *Circulation* 145 (24), 1749–1760. <https://doi.org/10.1161/CIRCULATIONAHA.121.057179> 35450432.
- Deng, W., Aimone, J.B., Gage, F.H., 2010. New neurons and new memories: how does adult hippocampal neurogenesis affect learning and memory? *Nat. Rev. Neurosci.* 11, 339–350.
- Diering, G.H., Huganir, R.L., 2018. The ampa receptor code of synaptic plasticity. *Neuron* 100, 314–329.
- Dinarello, C.A., 2018. Overview of the il-1 family in innate inflammation and acquired immunity. *Immunol. Rev.* 281, 8–27.
- Disdier, C., Chalansonnet, M., Gagnaire, F., Gaté, L., Cosnier, F., Devoy, J., et al., 2017. Brain inflammation, blood brain barrier dysfunction and neuronal synaptophysin decrease after inhalation exposure to titanium dioxide nano-aerosol in aging rats. *Sci. Rep.* 7, 12196.
- El Khayat, M., Halwani, D.A., Hneiny, L., Alameddine, I., Haidar, M.A., Habib, R.R., 2022. Impacts of climate change and heat stress on farmworkers' health: a scoping review. *Front. Public Health* 10, 782811.
- El-Husseini, A.E., Schnell, E., Chetkovich, D.M., Nicoll, R.A., Bredt, D.S., 2000. Pcd-95 involvement in maturation of excitatory synapses. *Science* 290, 1364–1368 (New York, N.Y.).
- Epstein, Y., Yanovich, R., 2019. Heatstroke. *N. Engl. J. Med.* 380, 2449–2459.
- Fiore, A.M., Naik, V., Leibensperger, E.M., 2015. Air quality and climate connections. *J. Air Waste Manag. Assoc.* 1995 (65), 645–685.
- Gao, Q., Zang, E., Bi, J., Dubrow, R., Lowe, S.R., Chen, H., et al., 2022. Long-term ozone exposure and cognitive impairment among Chinese older adults: a cohort study. *Environ. Int.* 160, 107072.
- Gaudel, A., Cooper, O.R., Chang, K.L., Bourgeois, I., Ziemke, J.R., Strode, S.A., et al., 2020. Aircraft observations since the 1990s reveal increases of tropospheric ozone at multiple locations across the Northern Hemisphere. *Sci. Adv.* 6.

- Glass, C.K., Saijo, K., Winner, B., Marchetto, M.C., Gage, F.H., 2010. Mechanisms underlying inflammation in neurodegeneration. *Cell* 140, 918–934.
- Greve, H.J., Dunbar, A.L., Lombo, C.G., Ahmed, C., Thang, M., Messenger, E.J., et al., 2022. The bidirectional lung brain-axis of amyloid- β pathology: ozone dysregulates the periplaque microenvironment. *Brain*, awac113 <https://doi.org/10.1093/brain/awac113> 35348636 (In press).
- Hanganu, A., Bedetti, C., Degroot, C., Mejia-Constain, B., Lafontaine, A.L., Soland, V., et al., 2014. Mild cognitive impairment is linked with faster rate of cortical thinning in patients with parkinson's disease longitudinally. *Brain* 137, 1120–1129.
- Haruwaka, K., Ikegami, A., Tachibana, Y., Ohno, N., Konishi, H., Hashimoto, A., et al., 2019. Dual microglia effects on blood brain barrier permeability induced by systemic inflammation. *Nat. Commun.* 10, 5816.
- Haykin, H., Rolls, A., 2021. The neuroimmune response during stress: a physiological perspective. *Immunology* 54, 1933–1947.
- Holm, S.M., Balmes, J.R., 2022. Systematic review of ozone effects on human lung function, 2013 through 2020. *Chest* 161, 190–201.
- Hou, L., Sun, F., Sun, W., Zhang, L., Wang, Q., 2019. Lesion of the locus coeruleus damages learning and memory performance in paraquat and maneb-induced mouse Parkinson's disease model. *Neuroscience* 419, 129–140.
- Huang, X., Hussain, B., Chang, J., 2021. Peripheral inflammation and blood-brain barrier disruption: effects and mechanisms. *CNS Neurosci. Ther.* 27, 36–47.
- Jhunjh, I., Fann, N., Zanolletti, A., Hubbell, B., 2014. Effect modification of ozone-related mortality risks by temperature in 97 US cities. *Environ. Int.* 73, 128–134.
- Kim, E., Sheng, M., 2004. Pd domain proteins of synapses. *Nat. Rev. Neurosci.* 5, 771–781.
- Kim, S., Kim, S.Y., Oh, J., Chae, Y., Park, J., Kim, D., et al., 2020. Effects of the 2018 heat wave on health in the elderly: implications for adaptation strategies to climate change. *Environ. Anal. Health Toxicol.* 35, e2020024–2020020.
- Kiyatkin, E.A., 2010. Brain temperature homeostasis: physiological fluctuations and pathological shifts. *Front. Biosci.* 15, 73–92 (Landmark edition).
- Lee, W., Moon, M., Kim, H.G., Lee, T.H., Oh, M.S., 2015. Heat stress-induced memory impairment is associated with neuroinflammation in mice. *J. Neuroinflammation* 12, 102.
- Lee, C.W., Hsu, L.F., Wu, I.L., Wang, Y.L., Chen, W.C., Liu, Y.J., et al., 2022. Exposure to polystyrene microplastics impairs hippocampus-dependent learning and memory in mice. *J. Hazard. Mater.* 430, 128431.
- Liu, F., Wang, Z., Wei, Y., Liu, R., Jiang, C., Gong, C., et al., 2021. The leading role of adsorbed lead in pm(2.5)-induced hippocampal neuronal apoptosis and synaptic damage. *J. Hazard. Mater.* 416, 125867.
- Loughnan, M.E., Nicholls, N., Tapper, N.J., 2010. When the heat is on: threshold temperatures for AMI admissions to hospital in Melbourne Australia. *Appl. Geogr.* 30, 63–69.
- Lyu, Y., Ju, Q., Lv, F., Feng, J., Pang, X., Li, X., 2022. Spatiotemporal variations of air pollutants and ozone prediction using machine learning algorithms in the Beijing-Tianjin-Hebei region from 2014 to 2021. *Environ. Pollut.* 306, 119420. <https://doi.org/10.1016/j.envpol.2022.119420> 35526642 (Barking, Essex: 1987).
- Moeller, D.R., Smith Jr., D.L., 2012. Ambient temperature and obesity. *Curr. Obes. Rep.* 1, 26–34.
- Nakamura, K., Nakamura, Y., Kataoka, N., 2022. A hypothalamomedullary network for physiological responses to environmental stresses. *Nat. Rev. Neurosci.* 23, 35–52.
- Nation, D.A., Sweeney, M.D., Montagne, A., Sagare, A.P., D'Orazio, L.M., Pachicano, M., et al., 2019. Blood-brain barrier breakdown is an early biomarker of human cognitive dysfunction. *Nat. Med.* 25, 270–276.
- National Research Council, 1991. Rethinking the Ozone Problem in Urban And Regional Air Pollution. The National Academies Press, Washington, DC.
- Nery-Flores, S.D., Mendoza-Magaña, M.L., Ramírez-Herrera, M.A., Ramírez-Vázquez, J.J., Romero-Prado, M.M.J., Cortez-Álvarez, C.R., et al., 2018. Curcumin exerted neuroprotection against ozone-induced oxidative damage and decreased nf- κ b activation in rat hippocampus and serum levels of inflammatory cytokines. *Oxidative Med. Cell. Longev.* 2018, 9620684.
- Niu, Y., Yang, T., Gu, X., Chen, R., Meng, X., Xu, J., et al., 2022. Long-term ozone exposure and small airway dysfunction: the China Pulmonary Health (CPH) study. *Am. J. Respir. Crit. Care Med.* 205, 450–458.
- Quinteiro-Filho, W.M., Rodrigues, M.V., Ribeiro, A., Ferraz-de-Paula, V., Pinheiro, M.L., Sá, L.R., et al., 2012. Acute heat stress impairs performance parameters and induces mild intestinal enteritis in broiler chickens: role of acute hypothalamic-pituitary-adrenal axis activation. *J. Anim. Sci.* 90, 1986–1994.
- Raza, A., Dahlquist, M., Lind, T., Ljungman, P.L.S., 2018. Susceptibility to short-term ozone exposure and cardiovascular and respiratory mortality by previous hospitalizations. *Environ. Health* 17, 37.
- Ronaldson, P.T., Davis, T.P., 2020. Regulation of blood-brain barrier integrity by microglia in health and disease: a therapeutic opportunity. *J. Cereb. Blood Flow Metab.* 40, S6–S24.
- Sinha, R.K., 2007. An approach to estimate EEG power spectrum as an index of heat stress using backpropagation artificial neural network. *Med. Eng. Phys.* 29, 120–124.
- Sun, G., Lin, X., Yi, X., Zhang, P., Liu, R., Fu, B., et al., 2021. Aircraft noise, like heat stress, causes cognitive impairments via similar mechanisms in male mice. *Chemosphere* 274, 129739.
- Sweeney, M.D., Sagare, A.P., Zlokovic, B.V., 2018. Blood-brain barrier breakdown in Alzheimer disease and other neurodegenerative disorders. *Nat. Rev. Neurol.* 14, 133–150.
- Tawatsupa, B., Lim, L.L., Kjellstrom, T., Seubsmann, S.A., Sleight, A., The Thai Cohort Study T, 2010. The association between overall health, psychological distress, and occupational heat stress among a large national cohort of 40,913 Thai workers. *Glob. Health Action* 3.
- Vorhees, C.V., Williams, M.T., 2006. Morris water maze: procedures for assessing spatial and related forms of learning and memory. *Nat. Protoc.* 1, 848–858.
- Wang, P., Chen, Y., Hu, J., Zhang, H., Ying, Q., 2019. Attribution of tropospheric ozone to NO (x) and VOC emissions: considering ozone formation in the transition regime. *Environ. Sci. Technol.* 53, 1404–1412.
- Welcome, M.O., Mastorakis, N.E., 2020. Stress-induced blood brain barrier disruption: molecular mechanisms and signaling pathways. *Pharmacol. Res.* 157, 104769.
- Wolf, H.K., Buslei, R., Schmidt-Kastner, R., Schmidt-Kastner, P.K., Pietsch, T., Wiestler, O.D., et al., 1996. Neun: a useful neuronal marker for diagnostic histopathology. *J. Histochem. Cytochem.* 44, 1167–1171.
- World Health Organization, 2017. Heatwaves. Available https://www.who.int/health-topics/heatwaves#tab=tab_1.
- World Health Organization, 2018. Heat and health. Available <https://www.who.int/news-room/fact-sheets/detail/climate-change-heat-and-health>.
- Yang, B.Y., Qian, Z., Howard, S.W., Vaughn, M.G., Fan, S.J., Liu, K.K., et al., 2018. Global association between ambient air pollution and blood pressure: a systematic review and meta-analysis. *Environ. Pollut.* 235, 576–588 (Barking, Essex: 1987).
- Yu, H., Ma, L., Liu, D., Wang, Y., Pei, X., Duan, Z., et al., 2020. Involvement of nmdar/psd-95/nmos-no-cgmp pathway in embryonic exposure to bpa induced learning and memory dysfunction of rats. *Environ. Pollut.* 266, 115055 (Barking, Essex: 1987).
- Zhang, D., Li, S., Hou, L., Jing, L., Ruan, Z., Peng, B., et al., 2021. Microglial activation contributes to cognitive impairments in rotenone-induced mouse Parkinson's disease model. *J. Neuroinflammation* 18, 4.
- Zhang, H., Wang, J., Ruan, C., Gao, Z., Zhu, Q., Li, S., 2022. Co-exposure of chronic stress and alumina nanoparticles aggravates hippocampal microglia pyroptosis by activating cathepsin b/nlrp3 signaling pathway. *J. Hazard. Mater.* 436, 129093.
- Zhong, J., Yu, H., Huang, C., Zhong, Q., Chen, Y., Xie, J., et al., 2018. Inhibition of phosphodiesterase 4 by fcpr16 protects sh-sy5y cells against mpp(+)-induced decline of mitochondrial membrane potential and oxidative stress. *Redox Biol.* 16, 47–58.
- Zhu, Q., Bi, J., Liu, X., Li, S., Wang, W., Zhao, Y., et al., 2022. Satellite-based long-term spatiotemporal patterns of surface ozone concentrations in China: 2005–2019. *Environ. Health Perspect.* 130, 27004.